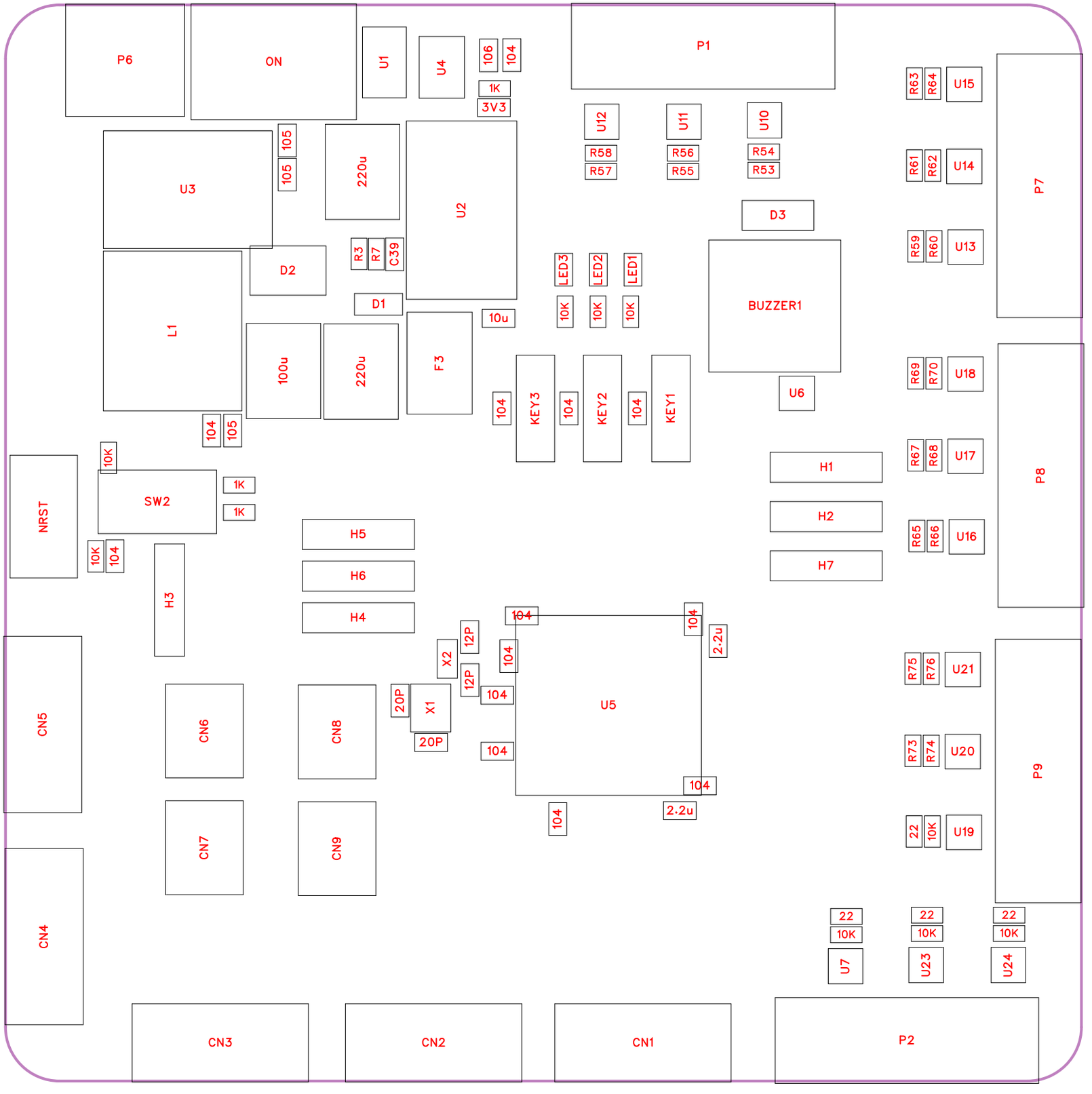

P6
ON
U1
U4
106
104
1K
3V3
P1
U12
U11
U10
R58
R57
R56
R55
R54
R53
R63
R64
U15
R61
R62
U14
R59
R60
U13
P7
U3
105
105
220u
U2
D3
BUZZER1
R3
R7
C39
D2
D1
LED3
LED2
LED1
10K
10K
10K
10u
L1
100u
220u
F3
104
KEY3
104
KEY2
104
KEY1
U6
104
105
10K
NRST
SW2
1K
1K
H1
H2
H7
R69
R70
U18
R67
R68
U17
R65
R66
U16
P8
H5
H6
H4
10K
104
H3
104
104
12P
X2
12P
104
104
U5
104
2.2u
CN5
CN6
CN8
20P
X1
20P
104
104
2.2u
R75
R76
U21
R73
R74
U20
22
10K
U19
P9
CN7
CN9
CN4
22
10K
U7
22
10K
U23
22
10K
U24
CN3
CN2
CN1
P2

No.	Quantity	Comment	Designator	Footprint	Value	Manufacturer Part	Manufacturer	Supplier Part	Supplier
1	6	1K	1K,1K,1K,10K,10K,10K	R0603	1K				
2	2	2.2uF	2.2u,2.2u	C0603	2.2uF				
3	4	XL-1608SURC-06	3V3,LED1,LED2,LED3	LED0603-RD_RED		XL-1608SURC-06	XINGLIGHT(成兴光)	C965799	LCSC
4	18	10K	10K,10K,10K,10K,10K,10K,R7,R54,R56,R58,R60,R62,R64,R66,R68,R70,R74,R76	R0603	10K				
5	1	10uF	10u	C0603	10uF	0603X106M250NT	FH(风华)	C479655	LCSC
6	2	12.5pF	12P,12P	C0603	12.5pF				
7	2	20pF	20P,20P	C0603	20pF	CL10C200JB8NNNC	SAMSUNG(三星)	C1648	LCSC
8	15	22R	22,22,22,22,R53,R55,R57,R59,R61,R63,R65,R67,R69,R73,R75	R0603	22R				
9	1	100uF	100u	CAP-SMD_BD6.3-L6.6-W6.6-LS7.3-FD-H7.7	100uF	JVJ50V100M6x8	JIERR(捷而瑞)	C46550415	LCSC
10	13	100nF	104,104,104,104,104,104,104,104,104,104,104,104,104	C0603	100nF				
11	1	1uF	105	C0603	1uF				
12	2	1uF	105,105	C0603	1uF	CL10A105KB8NNNC	SAMSUNG(三星)	C15849	LCSC
13	1	10uF	106	C0603	10uF				
14	2	220uF	220u,220u	CAP-SMD_BD6.3-L6.6-W6.6-FD	220uF	JVJ35V220M6x8	JIERR(捷而瑞)	C46550411	LCSC
15	1	2.4kHz	BUZZER1	BUZ-TH_BD12.0-P7.60-D0.6-FD	2.4kHz	TMB12A05	华能	C96093	LCSC
16	1	33nF	C39	C0603	33nF	CL10B333KB8NNNC	SAMSUNG(三星)	C21117	LCSC
17	5	WJ15EDGVC-3.81-04P-14-00A	CN1,CN2,CN3,CN4,CN5	CONN-TH_4P-P3.81_WJ2EDGVC-3.81-4P		WJ15EDGVC-3.81-04P-14-00A	KANGNEX(康奈克斯电气)	C8410	LCSC
18	4	WJ15EDGVC-3.81-02P-14-00A	CN6,CN7,CN8,CN9	CONN-TH_WJ15EDGVC-3.81-2P		WJ15EDGVC-3.81-02P-14-00A	KANGNEX(康奈克斯电气)	C8396	LCSC
19	1	SMF5.0A	D1	SOD-123FL_L2.7-W1.8-LS3.8-RD		SMF5.0A	R+O(宏嘉诚)	C19077497	LCSC
20	1	SB1045L	D2	TO-277_3P-L5.4-W4.0-LS6.5-TR		SB1045L	GOODWORK(固得沃克)	C908747	LCSC

21	1	SS310	D3	SMA_L4.3-W2.6-LS5.0-FD		SS310	R+O(宏嘉诚)	C7420364	LCSC
22	1	BSMD2920-600-16V	F3	F2920		BSMD2920-600-16V	BHFUSE(佰宏)	C910847	LCSC
23	7	PZ254V-11-04P	H1,H2,H3,H4,H5,H6,H7	HDR-TH_4P-P2.54-V-M		PZ254V-11-04P	XFCN(兴飞)	C2691448	LCSC
24	3	TS-1101-C-W	KEY1,KEY2,KEY3	SW-SMD_L6.0-W3.3-LS8.0		TS-1101-C-W	XKB Connectivity(中国星坤)	C318938	LCSC
25	1	47uH	L1	IND-SMD_L13.8-W12.6_H6.5	47uH	YSPI1365-470M	YJYCOIN(益嘉源)	C497913	LCSC
26	1	KH-6X6X5H-STM	NRST	SW-SMD_4P-L6.0-W6.0-P4.50-LS9.0_H5.0		KH-6X6X5H-STM	kinghelm(金航标)	C2837531	LCSC
27	1	HX KCD11-2P-R	ON	SW-TH_2P-L15.0-W10.5-P5.00_KCD11-2P-R		HX KCD11-2P-R	hanxia(韩下)	C6917318	LCSC
28	5	WJ15EDGVC-3.81-6P	P1,P2,P7,P8,P9	CONN-TH_6P_WJ15EDGVC-3.81-6P		WJ15EDGVC-3.81-6P	KANGNEX(康奈克斯电气)	C8415	LCSC
29	1	WJ500V-5.08-2P	P6	CONN-TH_2P-P5.00_WJ500V-5.08-2P		WJ500V-5.08-2P	KANGNEX(康奈克斯电气)	C8465	LCSC
30	1	3.3kΩ	R3	R0603	3.3kΩ	0603WAF3301T5E	UNI-ROYAL(厚声)	C22978	LCSC
31	1	DSIC02LS-P	SW2	SW-SMD_4P-L5.1-W6.1-P2.54-LS9.8		DSIC02LS-P	XKB Connection(中国星坤)	C489174	LCSC
32	1	P6SMB30CA	U1	SMB_L4.5-W3.6-LS5.2-BI		P6SMB30CA	R+O(宏嘉诚)	C19077601	LCSC
33	1	LM1085S-3.3/TR	U2	TO-263-3_L8.6-W10.2-P2.54-LS14.4-BR		LM1085S-3.3/TR	HGSEMI(华冠)	C444380	LCSC
34	1	XL4015E1	U3	TO-263-5_L10.2-W8.9-P1.70-TL		XL4015E1	XLSEMI(芯龙)	C51661	LCSC
35	1	BSMD1812-300-24V	U4	F1812		BSMD1812-300-24V	BHFUSE(佰宏)	C7500481	LCSC
36	1	STM32F407VET6	U5	LQFP-100_L14.0-W14.0-P0.50-LS16.0-BL		STM32F407VET6	ST(意法半导体)	C28730	LCSC
37	16	2N7002K	U6,U7,U10,U11,U12,U13,U14,U15,U16,U17,U18,U19,U20,U21,U23,U24	SOT-23-3_L2.9-W1.3-P0.95-LS2.4-BR		2N7002K	R+O(宏嘉诚)	C22379475	LCSC

38	1	12MHz	X1	CRYSTAL-SMD_4P-L3.2-W2.5-BL	12MHz	X322512MSB4SI	YXC(扬兴晶振)	C9002	LCSC
39	1	32.768kHz	X2	FC-135R_L3.2-W1.5	32.768kHz	Q13FC13500004	EPSON(爱普生)	C32346	LCSC

GND
ON
U1
U4
106
104
1K
3V3
EM1
U3
105
105
220u
D2
L1
D1
R7
R58
U2
10u
100u
220u
F3
U12
U11
U10
R57
R58
R55
R56
R53
R54
D3
BUZZER1
LED1
LED2
LED3
10K
10K
10K
KEY1
KEY2
KEY3
104
104
104
U6
DIR−
DIR+
STEP−
STEP+
EN−
EN+
R63
R64
U15
R61
R62
U14
R59
R60
U13
R69
R70
U18
R67
R68
U17
R65
R66
U16
GND
CLK
DIO
3V3
UART1
3V3
GND
T3_Tx
T3_Rx
NRST
10K
1
0
104
105
1K
1K
10K
104
U2_RX
U2_TX
GND
3V3
5V
3V3
GND
12P
104
X2
20P
X1
20P
12P
104
104
U5
104
2.2u
104
2.2u
104
24V
GND
EC5_A
EC5_B
K1
GND
K3
GND
K2
GND
K4
GND
24V
GND
EC4_A
EC4_B
24V
GND
EC3_A
EC3_B
24V
GND
EC2_A
EC2_B
24V
GND
EC1_A
EC1_B
R75
R76
U21
R73
R74
U20
22
10K
U19
22
10K
U7
22
10K
U23
22
10K
U24
DIR−
DIR+
STEP−
STEP+
EN−
EN+

